

AMENDMENT NO. 1 JUNE 2021

TO

IS 13593 : 1992 UPVC PIPE FITTINGS FOR USE WITH UPVC PIPES IN THE SUCTION AND DELIVERY LINES OF AGRICULTURAL PUMPS — SPECIFICATION

(Page 1, clause 1, Note) — Substitute ‘IS 12231’ for ‘IS 12231 : 1987’.

(Page 1, clause 2) — Substitute the following for the existing clause:

‘The standards listed below contain provisions which, through reference in this text, constitute provision of this standard. At the time of publication the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate, the possibility of applying the most recent editions of the standards.

<i>IS No.</i>	<i>Title</i>
554 : 1999/ ISO 7-1 : 1994	Pipe threads where pressure-tight joints are made on the threads — Dimensions, tolerances and designation (<i>fourth revision</i>)
1536 : 2001	Centrifugally cast (spun) iron pressure pipes for water, gas and sewage (<i>fourth revision</i>)
4905 : 2015/ ISO 24153 : 2009	Random sampling and randomization procedures (<i>first revision</i>)
4985 : 2021	Unplasticized PVC pipes for water supplies — Specification (<i>fourth revision</i>)

Price Group 2

Amendment No. 1 to IS 13593 : 1992

<i>IS No.</i>	<i>Title</i>
12231 : 1987	Specification for unplasticized PVC pipes for use in suction and delivery lines of agricultural pump sets
12235	Thermoplastics pipes and fittings — Methods of test (<i>first revision</i>)
(Part 4 : 2004)	Determining the detrimental effect on the composition of water
(Part 8/Sec 1 : 2004)	Resistance to internal hydraulic pressure, Section 1 Resistance to internal hydrostatic pressure at constant internal water pressure
(Part 10 : 2004)	Determination of organotin as tin aqueous solution'

(Page 2, clauses 4, 5, 7.1 and 7.2) — Substitute 'IS 12231' for 'IS 12231 : 1987'.

(Page 3, clause 7.4.2.1) — Substitute 'IS 554' for 'IS 554 : 1985'.

(Page 3, clauses 7.5) — Substitute 'IS 1536' for 'IS 1536 : 1989'.

(Page 3, clause 9.2) — Substitute 'IS 12235 (Part 8/Sec 1)' for 'IS 12235 (Part 8) : 1986' and 'IS 12231' for 'IS 12231 : 1987'.

(Page 3, clause 9.2) — Substitute 'IS 12235 (Part 8/Sec 1)' for 'IS 12235 (Part 8) : 1986' and 'IS 12231' for 'IS 12231 : 1987'.

(Page 3, clause 9.2) — Insert the following at the end:

‘9.3 Effect on Water

The fittings shall not have any detrimental effect on the composition of water flowing through them. When tested by the method described in IS 12235 (Part 4) and IS 12235 (Part 10), the quantities of lead, dialkyl tin C4 and higher homologues (measured as tin), and any other toxic substances extracted from

Amendment No. 1 to IS 13593 : 1992

the internal walls of the fittings shall not exceed the concentrations as specified in **10.3** of IS 4985 and meet the other requirements given in **10.3.1** of IS 4985. This test shall be taken as type test.

NOTE — Implementation of the phase out programme for use of lead stabilizers in PVC pipe and fitting manufacturing of the Government of India shall be borne in mind.’

(Page 4, *clause 10.2*) — Substitute ‘standard’ for ‘specification’.

(Page 4, *clause 10.2.1*) — Substitute ‘IS 4905’ for ‘IS 4905 : 1968’.

(Page 4, *clause 10.3.1*) — Substitute the following for the existing:

‘Each fitting so selected shall be examined for the specified requirements as given under **6.1.2**, **7.1**, **7.2**, **7.3**, **7.4**, **8** and **9.2** of this standard. Any fitting failing in one or more of these requirements shall be considered as defective fitting. The lot shall be deemed to have satisfied these requirements if the number of defective fittings found in the sample does not exceed the corresponding acceptance number, A given in *col 3* of Table 3.’

(Page 4, *clause 10.3.2*, line 9) — Substitute ‘standard’ for ‘specification’.

(Page 4, *clause 10.3.2*) — Insert the following at the end:

‘10.4 Type Tests

10.4.1 Type tests are intended to prove the suitability and performance of a new composition or a new size of pipe. Such tests, therefore, need to be applied only when a change is made in polymer composition or when a new size of pipe is to be introduced. Type tests for compliance with tests as specified in **9.1** and **9.3** shall be carried out.

10.4.2 Vacuum Test

10.4.2.1 For vacuum test, the manufacturer or the supplier shall furnish to the testing authority three samples of fittings of different diameters and different classes selected preferably from a regular production lot.

10.4.2.2 Three samples so selected shall be tested for compliance with the requirements for vacuum test as given in **9.1**.

Amendment No. 1 to IS 13593 : 1992

10.4.2.3 If all three samples pass the requirements for vacuum test, the type test of the pipe under consideration shall be considered to be eligible for approval, which shall be normally valid for a period of one year.

10.4.2.4 In case any of the samples fails in this test, the testing authority, at its discretion, may call for fresh samples not exceeding the original number, and subject them to vacuum test. If, in the repeat test, no single failure occurs, the type of pipe under consideration shall be considered eligible for type approval. If any of the samples fails in the repeat test, the type of pipe shall not be approved. The manufacturer or the supplier may be asked to improve the design and resubmit the product for type approval.

10.4.2.5 At the end of the validity period (normally one year) or earlier, if necessary, the testing authority may call for fresh samples for effect on water test for the purpose of type approval.

10.4.3 *Effect on Water*

10.4.3.1 For effect on water test, the manufacturer or the supplier shall furnish to the testing authority three samples of the smallest size of fitting selected preferably from a regular production lot.

10.4.3.2 Three samples so selected shall be tested for compliance with the requirements for effect on water as given in **9.3**.

10.4.2.3 If all three samples pass the requirements for effect on water, the type test of the fitting under consideration shall be considered to be eligible for approval, which shall be normally valid for a period of one year.

10.4.3.4 In case any of the samples fails in this test, the testing authority, at its discretion, may call for fresh samples not exceeding the original number, and subject them to the test for effect on water. If, in the repeat test, no single failure occurs, the type of fitting under consideration shall be considered eligible for type approval. If any of the samples fails in the repeat test, the type of fitting shall not be approved. The manufacturer or the supplier may be asked to improve the design and resubmit the product for type approval.

10.4.3.5 At the end of the validity period (normally one year) or earlier, if necessary, the testing authority may call for fresh samples for effect on water test for the purpose of type approval.'

Amendment No. 1 to IS 13593 : 1992

(Page 4, clause **11.1.1**) — Substitute the following for the existing:

‘**11.1.1** Lot number/Batch number shall be marked on the carton/packing and shall include the details of production in the following manner:

Year	Month	Day	Machine No.	Shift
XXXX	XX	XX	XXX	X

11.2 BIS Certification Marking

The fittings conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the fittings may be marked with the Standard Mark.’